

Sensing Security-Aware Beamforming for MIMO-OFDM ISAC via Deceptive AF Engineering

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Abstract—This work investigates enhancing sensing security in multi-input multi-output (MIMO) integrated sensing and communication (ISAC) systems in the presence of a passive sensing eavesdropper. To proactively impair unauthorized sensing, we exploit the deceptive ambiguity function (AF) engineering by introducing artificial sidelobes into the AF. To characterize the artificial-sidelobe-embedded AF, we propose a new metric, artificial sidelobe-to-mainlobe ratio (ASMR), and derive its closed-form expression with respect to the beamformer. Based on this analysis, a sensing security-aware MIMO beamforming problem is formulated and efficiently solved via an alternating optimization-based algorithm. Simulation results demonstrate that the proposed design effectively generates artificial ghost targets to confuse the eavesdropper while guaranteeing satisfactory ISAC performance.

Index Terms—Integrated sensing and communication (ISAC), sensing security, ambiguity function (AF), MIMO-OFDM.

I. INTRODUCTION

Integrated sensing and communication (ISAC) has emerged as a key technology for future 6G networks, aiming to realize simultaneous communication and sensing functionalities on one common platform through shared hardware and spectral resources [1]–[3]. Despite its spectral and functional benefits, ISAC also increases security risks due to the broadcast nature of wireless channels and spectrum sharing mechanism [2], [3].

Existing studies on secure ISAC primarily focus on communication security, which aims to protect data transmission against eavesdroppers by limiting communication information leakage [4]–[6] or hiding transmission existence [7]–[10]. Although the communication security-aware ISAC designs [4]–[9] effectively protect data transmission, they overlook the fact that ISAC signals can also be intercepted by unauthorized sensing receivers, causing sensitive sensing information to be inferred [11]. Facing this fact, some studies have investigated sensing security-aware ISAC designs [12]–[18]. To combat adversaries relying on power detection, early works focused on reducing the transmit power of ISAC signals [12], [13]. Furthermore, to combat advanced adversaries who may exploit feature extraction techniques like cyclostationary analysis to identify and characterize ISAC signals [11], the authors in [15] proposed low-probability-of-intercept signal design by

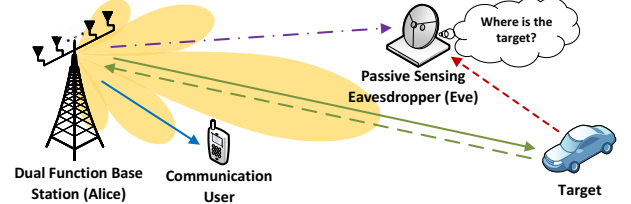


Fig. 1. Illustration of a MIMO-OFDM ISAC system in the presence of a passive sensing eavesdropper. The eavesdropper exploits the reference signal to demodulate the surveillance signal from the target.

minimizing the cyclic frequency sidelobe level of the cyclic spectrum. Although effective to some extent, the approaches [12]–[15] are defensive, primarily aiming to suppress sensing information leakage. To move beyond this passive protection, authors in [16] proposed an ambiguity-function (AF) engineering approach for sensing-secure orthogonal frequency division multiplexing (OFDM) ISAC by subcarrier power allocation, where artificial targets are injected into the range profile of a passive sensing eavesdropper by controlling peak and integrated sidelobe levels (PSL/ISL) of the squared autocorrelation function (ACF). However, this design is restricted to a single-antenna dual-function base station (DFBS), so the artificially induced sidelobes are intrinsic to the transmitted waveform. In this case, the legitimate receiver must suppress these sidelobes through reciprocal filtering, inevitably leading to sensing signal-to-noise ratio (SNR) loss. More importantly, such approaches become ineffective in multiple-input multiple-output (MIMO) systems, where sensing security is tightly coupled with spatial beamforming and cannot be controlled through subcarrier power allocation alone.

To overcome the aforementioned limitations, we investigate a sensing security-aware beamforming design for MIMO-OFDM ISAC systems. The contributions are listed as follows. *First*, we establish an analytical framework to characterize AF-based sensing deception and novelly introduce an artificial sidelobe-to-mainlobe ratio (ASMR) metric as an explicit function of the beamformer. *Second*, guided by this analysis, a sensing security-aware MIMO beamforming problem is formulated by maximizing the mainlobe power subject to deceptive AF shaping, communication rate, and power constraints. An efficient algorithm is proposed to solve this non-convex problem. Simulation results demonstrate that the proposed design effectively guarantees ISAC performance and improves sensing security by generating misleading ghost targets.

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II. SYSTEM MODEL

As shown in Fig. 1, we consider a MIMO-OFDM ISAC system, where a DFBS (Alice) equipped with M_A antennas probes a target of interest while simultaneously serving a communication user over K subcarriers. Besides, a legitimate sensing receiver collocated with Alice collects echoes reflected by the target for detection and estimation. There also exists a passive sensing eavesdropper (Eve) equipped with M_E antennas, who exploits the ISAC signals to determine the location of the target. Assume that both Alice and Eve adopt uniform linear arrays (ULAs) with half-wavelength element spacing.

A. Transmit Model

Let $\mathbf{s} = [s_1, \dots, s_K] \in \mathbb{C}^K$ denote the communication data symbols modeled as independent and identically distributed (i.i.d.) random variables uniformly drawn from the constellation set $\mathcal{S}$, assuming $\mathbb{E}[|s_k|^2] = 1$, $\mathbb{E}[|s_k|^4] = \mu_4$, $s_k \in \mathcal{S}$ [19]. The data symbol s_k is processed by a beamformer $\mathbf{f}_k \in \mathbb{C}^{M_A}$ in frequency domain and then transformed to the time domain via K -point inverse discrete Fourier transform (IDFT). After cyclic prefix (CP) addition, the signal is up-converted via radio frequency (RF) chains. Finally, the transmitted ISAC signal $\mathbf{x} \in \mathbb{C}^{M_A}$ emitted by M_A antennas at time instant $t \in (0, \Delta t]$ can be expressed as

$$\mathbf{x}(t) = \sum_{k=1}^K \mathbf{f}_k s_k e^{j2\pi(f_c + (k-1)\Delta f)t}, \quad (1)$$

where f_c is the center frequency. Δt and Δf denote the OFDM symbol duration and subcarrier spacing, respectively. They satisfy the subcarrier orthogonality requirement $\Delta t = 1/\Delta f$.

B. Communication Model

The transmitted signal (1) propagates through the mmWave communication channel and is received by the communication user. The received signal is firstly down-converted via RF chains and then transformed to frequency domain using K -point discrete Fourier transform (DFT) after removing the CP. The frequency-domain signal at the k -th subcarrier is given by

$$y_{C,k} = \mathbf{h}_k^H \mathbf{f}_k s_k + n_k, \forall k, \quad (2)$$

where $n_k \sim \mathcal{CN}(0, \sigma_C^2)$ and $\mathbf{h}_k \in \mathbb{C}^{M_A}$ denote the additive white Gaussian noise (AWGN) and CSI vector at the k -th subcarrier, respectively. Based on this, the achievable communication rate at the k -th subcarrier can be expressed as

$$R_k = \log_2 \left(1 + |\mathbf{h}_k^H \mathbf{f}_k|^2 / \sigma_C^2 \right), \forall k. \quad (3)$$

C. Alice's Reception Model

Suppose there is a target at $\theta_{A,T}$ from the perspective of Alice. The reflected signal received by the legitimate receiver is transformed to the frequency domain via DFT. The frequency-domain echo at the k -th subcarrier, $\mathbf{r}_{A,k} \in \mathbb{C}^{M_A}$, is given by

$$\begin{aligned} \mathbf{r}_{A,k} &= \alpha_{A,T} \mathbf{a}_A(\theta_{A,T}) \mathbf{a}_A^H(\theta_{A,T}) \mathbf{f}_k s_k e^{-j2\pi(k-1)\Delta f \tau_{A,T}} \\ &+ \alpha_{A,E} \mathbf{a}_A(\theta_{A,E}) \mathbf{a}_A^H(\theta_{A,E}) \mathbf{f}_k s_k e^{-j2\pi(k-1)\Delta f \tau_{A,E}} + \mathbf{z}_{A,k}, \end{aligned} \quad (4)$$

where $\alpha_{A,T}$ and $\alpha_{A,E}$ denotes the complex amplitude of the target and Eve. $\tau_{A,T}$ denotes the time-of-flight (ToF) from Alice to the target and back to Alice, and $\tau_{A,E}$ denotes the ToF from Alice to Eve and back to Alice. Besides, $\mathbf{z}_{A,k} \sim \mathcal{CN}(\mathbf{0}, \sigma_A^2 \mathbf{I}_{M_A})$ denotes the AWGN and $\mathbf{a}_A(\theta) =$

$(1/\sqrt{M_A})[1, e^{j\pi \sin \theta}, \dots, e^{j\pi(M_A-1) \sin \theta}]^T \in \mathbb{C}^{M_A}$ denotes the angular steering vector of Alice.

Then, the frequency-domain signal (4) is processed by the spatial filter using the steering vector $\mathbf{a}_A(\theta_{A,T})$ and the matched filter $\tilde{x}_{A,k} = (\mathbf{a}_A^H(\theta_{A,T}) \mathbf{f}_k s_k)^*$, $\forall k$. Specifically, the MF output signal can be expressed as

$$\begin{aligned} y_{A,MF,k} &= \alpha_{A,T} |\mathbf{a}_A^H(\theta_{A,T}) \mathbf{f}_k|^2 |s_k|^2 e^{-j2\pi(k-1)\Delta f \tau_{A,T}} \\ &+ \alpha_{A,E} \varsigma \mathbf{a}_A^H(\theta_{A,E}) \mathbf{f}_k \mathbf{a}_A^T(\theta_{A,T}) \mathbf{f}_k^* |s_k|^2 e^{-j2\pi(k-1)\Delta f \tau_{A,E}} \\ &+ \mathbf{a}_A^H(\theta_{A,T}) \mathbf{z}_{A,k} \mathbf{a}_A^T(\theta_{A,T}) \mathbf{f}_k^* s_k^*, \forall k. \end{aligned} \quad (5)$$

Based on (5), the output SINR at Alice is given by

$$\text{SINR}_A = \frac{P_{A,\text{signal}}}{P_{A,\text{interf}} + P_{A,\text{noise}}}, \quad (6)$$

where $P_{A,\text{signal}} = \mu_4 |\alpha_{A,T}|^2 \sum_{k=1}^K |\mathbf{a}_A^H(\theta_{A,T}) \mathbf{f}_k|^4$, $P_{A,\text{interf}} = \mu_4 |\alpha_{A,E}|^2 |\varsigma|^2 \sum_{k=1}^K |\mathbf{a}_A^H(\theta_{A,E}) \mathbf{f}_k|^2 |\mathbf{a}_A^H(\theta_{A,T}) \mathbf{f}_k|^2$, and $P_{A,\text{noise}} = \sigma_A^2 \sum_{k=1}^K |\mathbf{a}_A^H(\theta_{A,T}) \mathbf{f}_k|^2$, with defining $\varsigma = \mathbf{a}_A^H(\theta_{A,T}) \mathbf{a}_A(\theta_{A,E})$. From (5) and (6), we have the following remark.

Remark 1. Unlike the subcarrier power allocation design [16], where artificial sidelobes must be mitigated at Alice via reciprocal filtering with inevitable SNR loss, the proposed MIMO beamforming framework embeds sensing security in the spatial domain. As a result, reciprocal filtering is not required at Alice, and the associated SNR loss is completely avoided.

D. Eve's Reception Model

Note that Eve is a passive sensing receiver capable of detecting and estimating the target by exploiting ISAC signals. Specifically, Eve relies on two types of signals: 1) *Surveillance Signal* (Alice→Target→Eve) carries target information; 2) *Reference Signal* (Alice→Eve) is used to process the surveillance signal. The corresponding models are given as follows.

1) *Reference Signal Model*: Since Eve has no prior information about the transmitted signals, it needs the reference signal to demodulate the surveillance signal. Specifically, the reference signal at the k -th subcarrier is given by

$$\begin{aligned} \mathbf{r}_{E,\text{ref},k} &= \alpha_{E,\text{ref}} \mathbf{a}_E(\theta_{E,A}) \mathbf{a}_A^H(\theta_{A,E}) \mathbf{f}_k s_k e^{-j2\pi(k-1)\Delta f \tau_{E,\text{ref}}} \\ &+ \mathbf{z}_{E,\text{ref},k}, \end{aligned} \quad (7)$$

where $\alpha_{E,\text{ref}}$ and $\tau_{E,\text{ref}}$ denotes the complex amplitude and the ToF from Alice to Eve, respectively. Besides, $\mathbf{a}_E(\theta) = \frac{1}{\sqrt{M_E}}[1, e^{j\pi \sin \theta}, \dots, e^{j\pi(M_E-1) \sin \theta}]^T \in \mathbb{C}^{M_E}$ denotes the steering vector of Eve. $\mathbf{z}_{E,\text{ref},k} \sim \mathcal{CN}(\mathbf{0}, \sigma_{E,\text{ref}}^2 \mathbf{I}_{M_E})$ is the AWGN. Suppose Eve knows the location of Alice, the reference signal is processed by the spatial filter using the steering vector $\mathbf{a}_E(\theta_{E,A})$ and the range component $e^{-j2\pi(k-1)\Delta f \tau_{E,\text{ref}}}$ is removed. The reference signal can be further written as

$$\hat{\mathbf{r}}_{E,\text{ref},k} = \alpha_{E,\text{ref}} \mathbf{a}_A^H(\theta_{A,E}) \mathbf{f}_k s_k + \tilde{\mathbf{z}}_{E,\text{ref},k}, \quad (8)$$

where $\tilde{\mathbf{z}}_{E,\text{ref},k} = \mathbf{a}_E^H(\theta_{E,A}) \mathbf{z}_{E,\text{ref},k}$.

2) *Surveillance Signal Model*: The ISAC signal in (1) is reflected by the target and received by the passive bistatic sensing receiver, Eve. The resulting surveillance signal is weaker than the reference signal, which is given by

$$\begin{aligned} \mathbf{r}_{E,\text{sur},k} &= \alpha_{E,T} \mathbf{a}_E(\theta_{E,T}) \mathbf{a}_A^H(\theta_{A,T}) \mathbf{f}_k s_k e^{-j2\pi(k-1)\Delta f \tau_{E,T}} \\ &+ \mathbf{z}_{E,\text{sur},k} \end{aligned} \quad (9)$$

where $\mathbf{z}_{E,\text{sur},k} \sim \mathcal{CN}(\mathbf{0}, \sigma_{E,\text{sur}}^2 \mathbf{I}_{M_E})$ denotes the AWGN, and $\alpha_{E,T}$ and $\tau_{E,T}$ denote the complex amplitude and the ToF from Alice to the target and then to Eve.

After performing spatial filtering by scanning the entire angular domain, i.e., $\mathbf{a}_E(\theta_s), \theta_s \in [-90^\circ, 90^\circ]$, the surveillance signal is processed by matched filter based on the reference signal, $\tilde{r}_{E,\text{ref},k} = (\alpha_{E,\text{ref}} \mathbf{a}_A^H(\theta_{A,E}) \mathbf{f}_k s_k + \tilde{z}_{E,\text{ref},k})^*$, the resulting output can be expressed as

$$\begin{aligned} y_{E,\text{MF},k} &= \alpha_{E,\text{ref}}^* \mathbf{a}_E^H(\theta_s) \mathbf{z}_{E,\text{sur},k} \mathbf{a}_A^T(\theta_{A,E}) \mathbf{f}_k^* s_k^* \\ &+ \alpha_{E,T} \alpha_{E,\text{ref}}^* \chi \mathbf{a}_A^H(\theta_{A,T}) \mathbf{f}_k \mathbf{a}_A^T(\theta_{A,E}) \mathbf{f}_k^* |s_k|^2 e^{-j2\pi(k-1)\Delta f \tau_E} \\ &+ \alpha_{E,T} \chi \mathbf{a}_A^H(\theta_{A,T}) \mathbf{f}_k s_k e^{-j2\pi(k-1)\Delta f \tau_E} \tilde{z}_{E,\text{ref},k}^* \\ &+ \mathbf{a}_E^H(\theta_s) \mathbf{z}_{E,\text{sur},k} \tilde{z}_{E,\text{ref},k}^*, \forall k, \end{aligned}$$

where we define $\chi = \mathbf{a}_E^H(\theta_s) \mathbf{a}_E(\theta_{E,T})$. Note that $|\chi| \leq 1$ and the equality holds if and only if $\theta_s = \theta_{E,T}$. Now, the range profile can be obtained by applying IDFT to frequency-domain MF output signal across all the subcarriers, leading to

$$\begin{aligned} \mathbf{p}_{E,\text{MF}} &= \alpha_{E,T} \alpha_{E,\text{ref}}^* \chi^2 \mathbf{\Gamma}_K^H \mathbf{B}_1 \mathbf{B}_2 \mathbf{c}_1 + \alpha_{E,T} \chi \mathbf{\Gamma}_K^H \mathbf{B}_3 \mathbf{B}_2 \mathbf{c}_1 \\ &+ \alpha_{E,\text{ref}}^* \mathbf{\Gamma}_K^H \mathbf{B}_1 \hat{\mathbf{z}}_{E,\text{sur}} + \mathbf{\Gamma}_K^H \mathbf{B}_3 \hat{\mathbf{z}}_{E,\text{sur}}, \end{aligned} \quad (10)$$

where $\mathbf{\Gamma}_K \in \mathbb{C}^{K \times K}$ denotes the normalized DFT matrix, $\mathbf{B}_1 = \text{Diag}(\mathbf{a}_A^T(\theta_{A,E}) \mathbf{f}_1^* s_1^*, \dots, \mathbf{a}_A^T(\theta_{A,E}) \mathbf{f}_K^* s_K^*) \in \mathbb{C}^{K \times K}$, $\mathbf{B}_2 = \text{Diag}(1, e^{-j2\pi\Delta f \tau_E}, \dots, e^{-j2\pi(K-1)\Delta f \tau_E}) \in \mathbb{C}^{K \times K}$, $\mathbf{B}_3 = \text{Diag}(\tilde{z}_{E,\text{ref},1}^*, \dots, \tilde{z}_{E,\text{ref},K}^*) \in \mathbb{C}^{K \times K}$, $\mathbf{c}_1 = [\mathbf{a}_A^H(\theta_{A,T}) \mathbf{f}_1 s_1, \dots, \mathbf{a}_A^H(\theta_{A,T}) \mathbf{f}_K s_K]^T \in \mathbb{C}^K$, $\hat{\mathbf{z}}_{E,\text{sur}} = [\mathbf{a}_E^H(\theta_{E,T}) \mathbf{z}_{E,\text{sur},1}, \dots, \mathbf{a}_E^H(\theta_{E,T}) \mathbf{z}_{E,\text{sur},K}]^T \in \mathbb{C}^K$.

It is worth mentioning that the range profile (10) refers to the representation of the surveillance signal along the range dimension, and the peaks in the range profile correspond to potential targets at different propagation delays. To this end, the range profile (10) is critical for Eve to detect the target and estimate its location.

III. SENSING SECURITY PERFORMANCE ANALYSIS BASED ON AMBIGUITY FUNCTION

In this section, we analyze sensing security performance for the considered ISAC systems based on the range profile (10).

A. Ambiguity Function Derivation

Based on the range profile at Eve in (10), the output consists of a signal-dependent term and several noise-related terms. Since Alice can only design the transmitted ISAC signal, only the signal-dependent term is controllable, which corresponds to the first term in (10). This term is determined by the correlation property of the transmitted signal and can be characterized by the zero-Doppler cut of the AF, i.e., the ACF, defined as

$$\Lambda_E[n] = \frac{1}{K} \sum_{k=1}^K E_k e^{j2\pi \frac{n(k-1)}{K}}, \quad n = 0, \dots, K-1, \quad (11)$$

where we define $E_k = \mathbf{a}_A^T(\theta_{A,E}) \mathbf{f}_k^* \mathbf{a}_A^H(\theta_{A,T}) \mathbf{f}_k |s_k|^2, \forall k$.

B. Sensing Security Performance Metric

From Eve's perspective, target detection and localization rely on identifying dominant peaks in the range profile (10). Since the range profile structure is governed by the ACF of the transmitted ISAC signal, introducing additional peaks into the ACF produces artificial peaks in the range profile that are indistinguishable from true target responses. These artificial sidelobe peaks (ASPs) create ghost targets in the

range domain, thereby degrading Eve's detection and range estimation performance. To quantify this effect, we introduce artificial sidelobe-to-mainlobe ratio (ASMR), as defined below.

Definition 1. The ASMR is defined in the equation (12), where $\mathbf{A}_1 = \mathbf{a}_A(\theta_{A,T}) \mathbf{a}_A^H(\theta_{A,T})$, $\mathbf{A}_2 = \mathbf{a}_A(\theta_{A,E}) \mathbf{a}_A^H(\theta_{A,E})$, and $\mathcal{A}$ is the collection of the indexes for ASPs.

Based on **Definition 1**, we can give the following remark.

Remark 2. Compared with existing AF-based secure sensing designs [16], which rely on structured subcarrier power allocation to indirectly shape artificial sidelobes, the closed-form expression of ASMR with respect to beamformer allows sensing security to be directly formulated as tractable optimization constraints. This enhances analytical tractability and simplifies the optimization procedure, leading to greater design efficiency.

IV. DECEPTION-GENERATION-AIDED SECURE BEAMFORMING DESIGN

In this section, we first formulate the deception-generation-aided secure beamforming design problem and then solve it based on alternating optimization.

A. Problem Formulation

Guided by **Remark 2**, we propose a deception-generation-aided secure beamforming design that maximizes the mainlobe power towards the intended target while creating artificial target signatures to mislead Eve. Specifically, the design problem can be formulated as

$$\max_{\{\mathbf{f}_k\}} \sum_{k=1}^K |\mathbf{a}_A^H(\theta_{A,T}) \mathbf{f}_k|^2, \quad (13a)$$

$$\text{s.t. ASMR}_E(n_{\text{art}}) \geq \gamma, \forall n_{\text{art}} \in \mathcal{A}, \quad (13b)$$

$$R_k \geq \xi_k, \forall k, \quad (13c)$$

$$\|\mathbf{f}_k\|_F^2 \leq P_k, \forall k, \quad (13d)$$

where (13a) defines the mainlobe power as the objective function. Constraint (13b) enforces the ASMR limit γ for deception generation, while (13c) guarantees that the achievable rate at the k -th subcarrier meets the threshold ξ_k . Constraint (13d) imposes the transmit power limit P_k . Note that (13) is non-convex due to the quartic constraint with respect to $\mathbf{f}_k$.

B. Solutions to Problem (13)

To tackle this problem, we introduce auxiliary variables $\mathbf{w}_k = \mathbf{f}_k, \forall k$, reformulating the original problem (13) as

$$\min_{\{\mathbf{f}_k\}, \{\mathbf{w}_k\}} - \sum_{k=1}^K \mathbf{w}_k^H \mathbf{A}_1 \mathbf{w}_k, \quad (14a)$$

$$\begin{aligned} \text{s.t. } &\psi \sum_{k=1}^K \mathbf{w}_k^H \mathbf{A}_1 \mathbf{w}_k \mathbf{f}_k^H \mathbf{A}_2 \mathbf{f}_k \\ &+ \sum_{k=1}^K \sum_{l \neq k}^K c_{n_{\text{art}},k,l} \mathbf{w}_l^H \mathbf{A}_1 \mathbf{w}_k \mathbf{f}_k^H \mathbf{A}_2 \mathbf{f}_l \geq 0, \forall n_{\text{art}} \in \mathcal{A}, \end{aligned} \quad (14b)$$

$$|\mathbf{h}_k^H \mathbf{f}_k|^2 \geq \sigma_C^2 (2^{\xi_k} - 1), \forall k, \quad (14c)$$

$$\|\mathbf{f}_k\|_2^2 \leq P_k, \forall k, \quad (14d)$$

where $\rho > 0$ and $\{\mathbf{m}_k\}$ denote the penalty parameter and dual variable, respectively. Besides, we define $\psi = \mu_4 (1 - \gamma)$ and $c_{n_{\text{art}},k,l} = e^{j2\pi n_{\text{art}}(k-l)/K} - \gamma$.

Then, by applying first-order Taylor approximations [20]–[22], the non-convex objective in (14a) and the constraints in (14b)–(14c) are transformed into convex forms. Consequently,

$$\text{ASMR}_E(n_{\text{art}}) = \frac{\mathbb{E}[|\Lambda_E[n_{\text{art}}]|^2]}{\mathbb{E}[|\Lambda_E[0]|^2]} = \frac{\mu_4 \sum_{k=1}^K \mathbf{f}_k^H \mathbf{A}_1 \mathbf{f}_k \mathbf{f}_k^H \mathbf{A}_2 \mathbf{f}_k + \sum_{k=1}^K \sum_{l \neq k}^K \mathbf{f}_l^H \mathbf{A}_1 \mathbf{f}_k \mathbf{f}_k^H \mathbf{A}_2 \mathbf{f}_l e^{j2\pi n_{\text{art}}(k-l)/K}}{\mu_4 \sum_{k=1}^K \mathbf{f}_k^H \mathbf{A}_1 \mathbf{f}_k \mathbf{f}_k^H \mathbf{A}_2 \mathbf{f}_k + \sum_{k=1}^K \sum_{l \neq k}^K \mathbf{f}_l^H \mathbf{A}_1 \mathbf{f}_k \mathbf{f}_k^H \mathbf{A}_2 \mathbf{f}_l}, n_{\text{art}} \in \mathcal{A}, \quad (12)$$

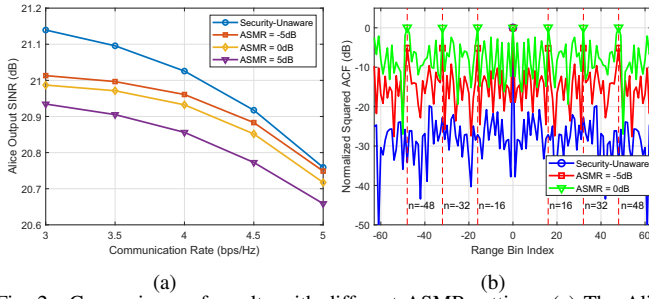


Fig. 2. Comparisons of results with different ASMR settings. (a) The Alice output SINR versus communication rate. (b) Normalized squared ACF.

problem (14) can be efficiently solved via an alternating optimization framework, as detailed in the following subproblems.

$$\min_{\{\mathbf{f}_k\}} \sum_{k=1}^K \|\mathbf{w}_k - \mathbf{f}_k + \mathbf{m}_k\|_F^2, \quad (15a)$$

$$\text{s.t. } g_{1,n_{\text{art}}} + 2 \sum_{k=1}^K \Re\{\bar{\mathbf{w}}_{n_{\text{art}},k}^H (\mathbf{f}_k - \bar{\mathbf{f}}_k)\} \geq 0, \forall n_{\text{art}} \in \mathcal{A}, \quad (15b)$$

$$\Xi_k \leq 2 \Re\{\bar{\mathbf{f}}_k^H \mathbf{h}_k \mathbf{h}_k^H \mathbf{f}_k\}, \forall k, \quad (15c)$$

$$\|\mathbf{f}_k\|_2^2 \leq P_k, \forall k, \quad (15d)$$

$$\min_{\{\mathbf{w}_k\}} -2 \sum_{k=1}^K \Re\{\bar{\mathbf{w}}_k^H \mathbf{A}_1 \mathbf{w}_k\} + \frac{\rho}{2} \sum_{k=1}^K \|\mathbf{w}_k - \mathbf{f}_k + \mathbf{m}_k\|_F^2, \quad (16a)$$

$$\text{s.t. } g_{2,n_{\text{art}}} + 2 \sum_{k=1}^K \Re\{\bar{\varphi}_{n_{\text{art}},k}^H (\mathbf{w}_k - \bar{\mathbf{w}}_k)\} \geq 0, \forall n_{\text{art}} \in \mathcal{A}, \quad (16b)$$

where

$$g_{1,n_{\text{art}}} = \sum_{k=1}^K \sum_{l \neq k}^K c_{n_{\text{art}},k,l} \mathbf{w}_l^H \mathbf{A}_1 \mathbf{w}_k \mathbf{f}_k^H \mathbf{A}_2 \mathbf{f}_l + \psi \sum_{k=1}^K \mathbf{w}_k^H \mathbf{A}_1 \mathbf{w}_k \mathbf{f}_k^H \mathbf{A}_2 \mathbf{f}_k,$$

$$\Xi_k = \sigma_C^2 (2^{\xi_k} - 1) + \bar{\mathbf{f}}_k^H \mathbf{h}_k \mathbf{h}_k^H \bar{\mathbf{f}}_k,$$

$$\bar{\mathbf{w}}_{n_{\text{art}},k} = \psi \mathbf{w}_k^H \mathbf{A}_1 \mathbf{w}_k \mathbf{A}_2 \bar{\mathbf{f}}_k + \frac{1}{2} (\sum_{l \neq k}^K c_{n_{\text{art}},k,l} \mathbf{w}_l^H \mathbf{A}_1 \mathbf{w}_k \mathbf{A}_2 \bar{\mathbf{f}}_l + \sum_{l \neq k}^K c_{n_{\text{art}},l,k} \mathbf{w}_k^H \mathbf{A}_1 \mathbf{w}_l \mathbf{A}_2 \bar{\mathbf{f}}_l),$$

$$\bar{\varphi}_{n_{\text{art}},k} = \psi \mathbf{f}_k^H \mathbf{A}_2 \mathbf{f}_k \mathbf{A}_1 \bar{\mathbf{w}}_k + \frac{1}{2} (\sum_{l \neq k}^K c_{n_{\text{art}},l,k} \mathbf{f}_l^H \mathbf{A}_2 \mathbf{f}_k \mathbf{A}_1 \bar{\mathbf{w}}_l + \sum_{l \neq k}^K c_{n_{\text{art}},k,l} \mathbf{f}_k^H \mathbf{A}_2 \mathbf{f}_l \mathbf{A}_1 \bar{\mathbf{w}}_l),$$

$$g_{2,n_{\text{art}}} = \sum_{k=1}^K \sum_{l \neq k}^K c_{n_{\text{art}},k,l} \mathbf{f}_k^H \mathbf{A}_2 \mathbf{f}_l \mathbf{w}_l^H \mathbf{A}_1 \mathbf{w}_k + \psi \sum_{k=1}^K \mathbf{f}_k^H \mathbf{A}_2 \mathbf{f}_k \mathbf{w}_k^H \mathbf{A}_1 \mathbf{w}_k,$$

$\bar{\mathbf{x}}$ denotes the last iteration point of $\mathbf{x}$. Note that subproblems (15) and (16) are convex and can be solved by existing solvers.

V. NUMERICAL SIMULATIONS

In this section, we provide some simulation results to evaluate the proposed sensing security-aware beamforming design.

A. Simulation Setup

Unless otherwise specified, Alice and Eve are equipped with $M_A = 16$ and $M_E = 64$ antennas, respectively. An OFDM system with $K = 64$ subcarriers, center frequency $f_c = 40$ GHz, bandwidth $B = 50$ MHz, and $N = 32$ coherent OFDM symbols is considered. The transmit power budget is $P_k = 30$ dBm, $\forall k$. The communication noise power is $\sigma_C^2 = 20$ dBm, while the sensing noise powers at Alice and Eve are $\sigma_A^2 = \sigma_E^2 = 0$ dBm. 16QAM modulation is adopted, and the ASP index set is $\mathcal{A} = 16, 32, 48$. For demonstration convenience, the target ranges observed by Alice and Eve are assumed identical.

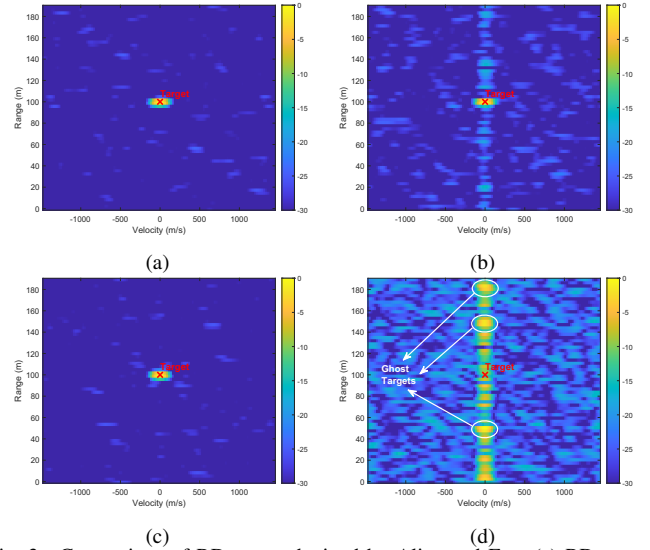


Fig. 3. Comparison of RD maps obtained by Alice and Eve. (a) RD map at Alice if security-unaware; (b) RD map at Eve if security-unaware; (c) RD map at Alice if security-aware; (d) RD map at Eve if security-aware.

B. Simulation Results

Fig. 2 illustrates the impact of different ASMR settings. As shown in Fig. 2(a), Alice's output SINR decreases with increasing communication rate, revealing a tradeoff among communication, sensing, and security. The security-unaware design achieves the highest SINR, while the case with ASMR = 5dB yields the lowest. Fig. 2(b) depicts the normalized squared ACF, where clear ASPs are generated at the desired range bins $n = \pm 16, \pm 32, \pm 48$ under ASMR = 0dB and -5dB. These results confirm the effectiveness of the proposed design.

Fig. 3 compares the RD maps observed by Alice and Eve, with ASMR = 0dB and $\xi_k = 3$. Under the sensing security-unaware design (Figs. 3(a) and 3(b)), both Alice and Eve exhibit a dominant peak at the true target location. Under the sensing security-aware design (Figs. 3(c) and 3(d)), Alice still observes a clear peak at the true target, whereas Eve observes multiple additional peaks at other locations. This indicates that the proposed deception-generation-aided beamforming successfully creates "ghost targets" to mislead Eve, thereby enhancing the sensing security of ISAC systems.

VI. CONCLUSION

In this paper, we focused on enhancing sensing security for MIMO-OFDM ISAC systems. To confuse a passive sensing eavesdropper, we exploited the deceptive AF engineering approach to generate artificial targets in the range profile of the eavesdropper. Driven by this, we made a sensing security analysis and introduced a new metric, ASMR, to characterize the artificial-sidelobe-embedded AF. Moreover, we formulated a sensing security-aware beamforming design problem by maximizing the mainlobe power subject to deceptive AF shaping, communication and power constraints, and proposed an effective algorithm to solve it. Simulation results validate the effectiveness and feasibility of the proposed design.

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